

Energy Approach for Global Existence of Some 1-D Quasilinear Hyperbolic Systems: Initial-boundary Value Problem Case

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Abstract

For the initial-boundary value problem of some one-dimensional quasilinear hyperbolic systems of diagonal form on $\mathbb{R}^+ \times \mathbb{R}^+$, we prove the global existence of classical solutions with small initial data, using some weighted energy approach.

keywords: Quasilinear hyperbolic systems; global existence.

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1 Main result

In this manuscript, we will consider the following quasilinear hyperbolic system:

$$\partial_t u_i + \lambda_i \partial_x u_i = \sum_{j=1}^n \lambda_{ij} u_j \partial_x u_i + \sum_{j,k=1}^n \lambda_{ijk} u_j u_k, \quad i = 1, 2, \dots, n, \quad (1.1)$$

where $n \geq 2$, λ_i ($i = 1, 2, \dots, n$) are some different nonzero constants, λ_{ij} and λ_{ijk} ($i, j, k = 1, 2, \dots, n$) are some constants, and $u = (u_1, \dots, u_n)$ is the unknown vector-valued function of (t, x) . We also further always assume

$$\lambda_1 < \dots < \lambda_p < 0 < \lambda_{p+1} < \dots < \lambda_n \quad (1.2)$$

with $1 \leq p < n$.

In the Cauchy problem case, i.e., for the system (1.1) on $\mathbb{R}^+ \times \mathbb{R}$, in [9], the first author proved that if there is no self-interaction in the nonlinear terms, that is,

$$\lambda_{ii} = 0, \quad i = 1, \dots, n, \quad (1.3)$$

and

$$\lambda_{ijj} = 0, \quad i, j = 1, \dots, n, \quad (1.4)$$

then it admits a unique global classical solution with small initial data, by using some weighted energy estimates inspired by [7], [1] and [6].

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The aim of this manuscript is to extend the above global existence result from the Cauchy problem case to the initial-boundary value problem case. Specifically speaking, we will consider the system (1.1) on $\mathbb{R}^+ \times \mathbb{R}^+$ with homogeneous boundary condition

$$u_i(t, 0) = 0, \quad i = p+1, \dots, n, \quad (1.5)$$

and initial data

$$t = 0 : u = f(x). \quad (1.6)$$

In view of (1.2), we know that there exist positive constants δ_0 and c such that

$$\lambda_{i+1} - \lambda_i \geq \delta_0, \quad i = 1, \dots, n-1, \quad (1.7)$$

and

$$c^{-1} \langle x - \lambda_i t \rangle \leq \langle x - \lambda_p t \rangle \leq \dots \leq \langle x - \lambda_1 t \rangle, \quad i = p+1, \dots, n, \quad (1.8)$$

on $\mathbb{R}^+ \times \mathbb{R}^+$.

In order to ensure the global existence, we will assume

$$\lambda_{ii} = 0, \quad i = p+1, \dots, n, \quad (1.9)$$

$$\lambda_{ijj} = 0, \quad i = 1, \dots, p, \quad j = i+1, \dots, n, \quad (1.10)$$

and

$$\lambda_{ijj} = 0, \quad i = p+1, \dots, n, \quad j = p+1, \dots, n. \quad (1.11)$$

The main result in this manuscript is the following:

Theorem 1.1. *Assume that (1.2), (1.9), (1.10) and (1.11) are satisfied. For all $0 < \delta < 1$, there exist a positive constant ϵ_0 such that for any $0 < \epsilon \leq \epsilon_0$, if*

$$\sum_{l=0}^2 \|\langle x \rangle^{1+\delta} \partial_x^l f\|_{L^2(\mathbb{R}^+)} \leq \epsilon \quad (1.12)$$

with $\langle \cdot \rangle = (1 + |\cdot|^2)^{1/2}$, then the initial-boundary value problem (1.1), (1.5), (1.6) admits a unique global classical solution.

Remark 1.1. *Compared (1.9), (1.10) and (1.11) with (1.3) and (1.4), we can see that in the initial-boundary value problem case, less restriction is needed in the interaction of nonlinear terms.*

1.1 Previous Results

For the Cauchy problem of semilinear wave equations with null conditions in one spatial dimension, the global existence of classical solutions with small initial data was established by [7] (an earlier result appears in [8]). This can be seen as a one-dimensional counterpart to the seminal works of [2] and [4] in three dimensions, and of [1] in two dimensions, on global existence for nonlinear wave equations under null conditions.

However, the mechanism responsible for global existence in one dimension differs fundamentally from those in higher dimensions. In the three- and two-dimensional proofs of [2], [4], and [1], the decay of linear waves plays a key role—a property absent in one spatial dimension, where waves do not decay. Instead, the one-dimensional global existence relies on the interaction of waves with different speeds, which induces decay in the nonlinear terms. To capture this interaction, [7] introduced a weighted energy estimate with suitably chosen positive weights.

In [9], the first author demonstrated that the positive-weight energy estimate introduced in [7], combined with a space-time weighted energy method inspired by [1] (and [6]), can be applied to establish the global existence of classical solutions for the Cauchy problem of certain one-dimensional diagonal quasilinear hyperbolic systems without self-interaction, under small initial data.

The characteristic method is well established as a powerful tool for one-dimensional quasilinear hyperbolic systems, with a comprehensive treatment provided in the monograph [5]. However, this approach is generally not applicable to multidimensional systems, where the energy method remains the only robust framework. Importantly, insights gained from the one-dimensional energy analysis can still inform the study of higher-dimensional problems. For instance, system (1.1) can be interpreted as a model for incompressible magnetohydrodynamics (MHD) under a strong background magnetic field (see [3]). We therefore expect that the results presented in this work may contribute to the treatment of initial-boundary value problems for MHD in a half-space.

2 Preliminaries

Denote

$$\phi(x) = \langle x \rangle^{2+2\delta}. \quad (2.1)$$

It is easy to verify that

$$|\phi'(x)| \leq 4\langle x \rangle^{1+2\delta}. \quad (2.2)$$

Denote

$$q(\rho) = \int_{-\infty}^{\rho} \langle s \rangle^{-(1+\delta)} ds \quad (2.3)$$

and

$$\psi_{ik}(x) = \exp(q((\lambda_k - \lambda_i)x)). \quad (2.4)$$

We can verify that

$$\psi'_{ik}(x) = (\lambda_k - \lambda_i)\psi_{ik}(x)\langle(\lambda_k - \lambda_i)x\rangle^{-(1+\delta)}. \quad (2.5)$$

We note that there exists a positive constant c_1 such that

$$c_1^{-1} \leq \psi_{ik}(x) \leq c_1, \quad (2.6)$$

and noting (1.7), we also have that there exists a positive constant c_2 such that

$$c_2^{-1}\langle x \rangle^{-(1+\delta)} \leq (\lambda_k - \lambda_i)\psi'_{ik}(x) \leq c_2\langle x \rangle^{-(1+\delta)}, \quad (2.7)$$

for any $k \neq i$.

Inspired by [7] and following [9], for $0 < \delta < 1$, we denote the weighted energy with positive weight by

$$E(u(t)) = \sum_{i=1}^n \|\langle x - \lambda_i t \rangle^{1+\delta} u_i(t, x)\|_{L_x^2(\mathbb{R}^+)}^2, \quad (2.8)$$

and the corresponding k -th ($k = 0, 1, 2, \dots$) order energy by

$$E_k(u(t)) = \sum_{l=0}^k E(\partial_{t,x}^l u(t)). \quad (2.9)$$

Inspired by [1], as in [9] we also use the following space-time weighted energy

$$\mathcal{E}(u(t)) = \sum_{i=1}^n \sum_{j \neq i} \|\langle x - \lambda_j s \rangle^{\frac{-1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} u_i(s, x)\|_{L_{s,x}^2(S_t)}^2, \quad (2.10)$$

in which $S_t = [0, t] \times \mathbb{R}^+$, and the corresponding k -th order version

$$\mathcal{E}_k(u(t)) = \sum_{l=0}^k \mathcal{E}(\partial_{t,x}^l u(t)). \quad (2.11)$$

We also need some notations concerning boundary terms. We denote

$$P^-(u(t)) = \sum_{i=1}^p \|\langle s \rangle^{1+\delta} u_i(s, 0)\|_{L_s^2(0,t)}^2 \quad (2.12)$$

and the corresponding k -th order version

$$P_k^-(u(t)) = \sum_{l=0}^k P^-(\partial_{t,x}^l u(t)); \quad (2.13)$$

and use

$$P^+(u(t)) = \sum_{i=p+1}^n \|\langle s \rangle^{1+\delta} u_i(s, 0)\|_{L_s^2(0,t)}^2 \quad (2.14)$$

and the corresponding k -th order version

$$P_k^+(u(t)) = \sum_{l=0}^k P^+(\partial_{t,x}^l u(t)). \quad (2.15)$$

The following two lemmas are key in the proof of Theorem 1.1.

Lemma 2.1. *Consider the following linear hyperbolic system*

$$\partial_t v_i + \lambda_i \partial_x v_i = \sum_{j=i}^n \lambda_{ij} u_j \partial_x v_i + F_i, \quad i = 1, 2, \dots, n \quad (2.16)$$

on $\mathbb{R}^+ \times \mathbb{R}^+$, where $u = (u_1, \dots, u_n)$ and $F = (F_1, \dots, F_n)$ are some given vector-valued functions of $(t, x) \in \mathbb{R}^+ \times \mathbb{R}^+$. Assume that (1.2) and (1.9) are satisfied and

$$\epsilon_1 = \sup_{0 \leq s \leq t} E_1^{1/2}(u(s)) \quad (2.17)$$

is sufficiently small. Then it holds that

$$P^-(v(t)) \leq CE(v(0)) + C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}(v(t))$$

$$+ C \sum_{i=1}^p \int_0^t \|\langle x - \lambda_i s \rangle^{2+2\delta} v_i F_i\|_{L_x^1(\mathbb{R}^+)} ds \quad (2.18)$$

and

$$\begin{aligned} & \sup_{0 \leq s \leq t} E(v(s)) + \mathcal{E}(v(t)) \\ & \leq CE(v(0)) + CP^+(v(t)) + C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}(v(t)) \\ & + C \sum_{i=1}^n \int_0^t \|\langle x - \lambda_i s \rangle^{2+2\delta} v_i F_i\|_{L_x^1(\mathbb{R}^+)} ds. \end{aligned} \quad (2.19)$$

Proof. First, it follows from Sobolev embedding $H^1(\mathbb{R}^+) \hookrightarrow L^\infty(\mathbb{R}^+)$ that

$$\sup_{0 \leq s \leq t} \|u(s, x)\|_{L_x^\infty(\mathbb{R}^+)} \leq C \sup_{0 \leq s \leq t} E_1^{1/2}(u(s)) \leq C\epsilon_1 \quad (2.20)$$

is sufficiently small.

Consider the i -th equation

$$\partial_t v_i + \lambda_i \partial_x v_i = \sum_{j=1}^n \lambda_{ij} u_j \partial_x v_i + F_i. \quad (2.21)$$

For $1 \leq k \leq n$, $k \neq i$, multiply $2\psi_{ik}(x - \lambda_k t)\phi(x - \lambda_i t)v_i$ on both sides of (2.21). Then Leibniz's rule gives

$$\begin{aligned} & (\partial_t + \lambda_i \partial_x)(\psi_{ik}(x - \lambda_k t)\phi(x - \lambda_i t)v_i^2) + (\lambda_k - \lambda_i)\psi'_{ik}(x - \lambda_k t)\phi(x - \lambda_i t)v_i^2 \\ & = \sum_{j=1}^n \partial_x(\psi_{ik}(x - \lambda_k t)\phi(x - \lambda_i t)\lambda_{ij}u_j v_i^2) - \sum_{j=1}^n \psi_{ik}(x - \lambda_k t)\phi(x - \lambda_i t)\lambda_{ij}(\partial_x u_j)v_i^2 \\ & \quad - \sum_{j=1}^n (\psi'_{ik}(x - \lambda_k t)\phi_{ik}(x - \lambda_i t) + \psi_{ik}(x - \lambda_k t)\phi'(x - \lambda_i t))\lambda_{ij}u_j v_i^2 \\ & \quad + 2\psi_{ik}(x - \lambda_k t)\phi(x - \lambda_i t)v_i F_i. \end{aligned} \quad (2.22)$$

The combination of (2.22), the fundamental theorem of calculus, (2.2), (2.6) and (2.7) implies

$$\begin{aligned} & \int_{\mathbb{R}^+} \langle x - \lambda_i t \rangle^{2+2\delta} v_i^2 dx + c_2^{-1} \sum_{k \neq i} \int_0^t \int_{\mathbb{R}^+} \langle x - \lambda_k s \rangle^{-(1+\delta)} \langle x - \lambda_i s \rangle^{2+2\delta} v_i^2 dx ds \\ & + \sum_{k \neq i} \int_0^t (-\lambda_i + \sum_{j=1}^n \lambda_{ij} u_j(s, 0)) \psi(-\lambda_k s) \phi(-\lambda_i s) v_i^2(s, 0) ds \\ & \leq C \int_{\mathbb{R}^+} \langle x \rangle^{2+2\delta} v_i^2(0, x) dx + C \sum_{j=1}^n \int_0^t \int_{\mathbb{R}^+} |\lambda_{ij}| \langle x - \lambda_i s \rangle^{2+2\delta} (|\partial_x u_j| + |u_j|) v_i^2 dx ds \\ & + C \int_0^t \int_{\mathbb{R}^+} \langle x - \lambda_i s \rangle^{2+2\delta} |v_i| |F_i| dx ds. \end{aligned} \quad (2.23)$$

We first consider the case $1 \leq i \leq p$. In this case, because $\lambda_i < 0$ and u is sufficiently small, for the third term on the left hand side of (2.23), noting (2.6), we have

$$\sum_{k \neq i} \int_0^t (-\lambda_i + \sum_{j=1}^n \lambda_{ij} u_j(s, 0)) \psi_{ik}(-\lambda_k s) \phi(-\lambda_i s) v_i^2(s, 0) ds$$

$$\geq C \|\langle s \rangle^{1+\delta} v_i(s, 0)\|_{L^2_s(0,t)}^2. \quad (2.24)$$

Now we need to estimate the second term on the right hand side of (2.23). It follows from Hölder inequality and Sobolev embedding $H^1(\mathbb{R}^+) \hookrightarrow L^\infty(\mathbb{R}^+)$ that

$$\begin{aligned} & \sum_{j \neq i} \|\langle x - \lambda_i s \rangle^{2+2\delta} (|\partial_x u_j| + |u_j|) v_i^2\|_{L^1_{t,x}} \\ & \leq \sum_{j \neq i} \|\langle x - \lambda_j s \rangle^{\frac{-1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} v_i\|_{L^2_{t,x}}^2 \|\langle x - \lambda_j s \rangle^{1+\delta} (|\partial_x u_j| + |u_j|)\|_{L^\infty_{t,x}} \\ & \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}(v(t)). \end{aligned} \quad (2.25)$$

Noting (1.8), we also have

$$\begin{aligned} & \|\langle x - \lambda_i s \rangle^{2+2\delta} (|\partial_x u_i| + |u_i|) v_i^2\|_{L^1_{t,x}} \\ & \leq \|\langle x - \lambda_n s \rangle^{\frac{-1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} v_i\|_{L^2_{t,x}}^2 \|\langle x - \lambda_n s \rangle^{1+\delta} (|\partial_x u_i| + |u_i|)\|_{L^\infty_{t,x}} \\ & \leq C \|\langle x - \lambda_n s \rangle^{\frac{-1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} v_i\|_{L^2_{t,x}}^2 \|\langle x - \lambda_i s \rangle^{1+\delta} (|\partial_x u_i| + |u_i|)\|_{L^\infty_{t,x}} \\ & \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}(v(t)). \end{aligned} \quad (2.26)$$

Thus it follows from (2.23)–(2.26) that (2.18) holds.

Now we consider the case $p+1 \leq i \leq n$. In this case, for the third term on the left hand side of (2.23), we have

$$\begin{aligned} & \sum_{k \neq i} \int_0^t \left(-\lambda_i + \sum_{j=1}^n \lambda_{ij} u_j(s, 0) \right) \psi(-\lambda_k s) \phi(-\lambda_i s) v_i^2(s, 0) ds \\ & \geq -C \|\langle s \rangle^{1+\delta} v_i(s, 0)\|_{L^2_s(0,t)}^2. \end{aligned} \quad (2.27)$$

For the second term on the right hand side of (2.23), noting (1.9), we can estimate it as (2.25).

From the above discussion, we can see that (2.19) holds. $\square$

In order to apply Lemma 2.1 to show Theorem 1.1, we still need to control the boundary terms concerning the positive characteristics. The following lemma provides such estimate.

Lemma 2.2. *Let u be a classical solution to (1.1) on $\mathbb{R}^+ \times \mathbb{R}^+$ with homogeneous boundary condition (1.5). Assume that (1.2) is satisfied and*

$$\epsilon_2 = \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \quad (2.28)$$

is sufficiently small. Then we have

$$P_2^+(u(t)) \leq C P_2^-(u(t)). \quad (2.29)$$

Proof. Noting that the time derivative preserves the homogeneous boundary condition, we have

$$P_2^+(u(t)) = P^+(\partial_x u(t)) + P^+(\partial_x \partial_t u(t)) + P^+(\partial_t^2 u(t)). \quad (2.30)$$

First, it follows from (1.1) that for $i = p+1, \dots, n$,

$$(\lambda_i - \sum_{j=1}^n \lambda_{ij} u_j) \partial_x u_i = -\partial_t u_i + \sum_{j,k=1}^n \lambda_{ijk} u_j u_k. \quad (2.31)$$

In view of (1.2), and noting that $|u|$ is sufficiently small, we have

$$|\partial_x u_i| \leq C |\partial_t u_i| + C \sum_{j,k=1}^n |\lambda_{ijk} u_j u_k|. \quad (2.32)$$

Furthermore, by (2.32) and (1.5) we get

$$|\partial_x u_i(t, 0)| \leq C \sum_{j,k=1}^p |u_j(t, 0) u_k(t, 0)|, \quad (2.33)$$

which implies

$$\begin{aligned} \|\langle s \rangle^{1+\delta} \partial_x u_i(s, 0)\|_{L_s^2(0,t)} &\leq C \sum_{j,k=1}^p \|\langle s \rangle^{1+\delta} u_j(s, 0)\|_{L_s^2(0,t)} \|u_k(s, 0)\|_{L_s^\infty(0,t)} \\ &\leq C \sum_{j=1}^p \|\langle s \rangle^{1+\delta} u_j(s, 0)\|_{L_s^2(0,t)} \sup_{0 \leq s \leq t} E_1^{1/2}(u(s)). \end{aligned} \quad (2.34)$$

Thus we have

$$P^+(\partial_x u(t)) \leq C P^-(u(t)). \quad (2.35)$$

Now we consider the second order terms. From (1.1) we have that for $i = p+1, \dots, n$,

$$\begin{aligned} (\lambda_i - \sum_{j=1}^n \lambda_{ij} u_j) \partial_x \partial_t u_i &= -\partial_t^2 u_i + \sum_{j=1}^n \lambda_{ij} \partial_t u_j \partial_x u_i \\ &\quad + \sum_{j,k=1}^n \lambda_{ijk} u_j \partial_t u_k + \sum_{j,k=1}^n \lambda_{ijk} \partial_t u_j u_k. \end{aligned} \quad (2.36)$$

In view of (1.2), and noting that $|u|$ is sufficiently small, we have

$$|\partial_x \partial_t u_i| \leq C |\partial_t^2 u_i| + C \sum_{j=1}^n |\lambda_{ij} \partial_t u_j \partial_x u_i| + C \sum_{j,k=1}^n |\lambda_{ijk} u_j \partial_t u_k| + C \sum_{j,k=1}^n |\lambda_{ijk} \partial_t u_j u_k|. \quad (2.37)$$

By (2.37) and (1.5) we get

$$|\partial_x \partial_t u_i(t, 0)| \leq C \sum_{j=1}^p |\partial_t u_j(t, 0) \partial_x u_i(t, 0)| + C \sum_{j,k=1}^p |\partial_t u_j(t, 0) u_k(t, 0)|, \quad (2.38)$$

which implies

$$\begin{aligned} &\|\langle s \rangle^{1+\delta} \partial_x \partial_t u_i(s, 0)\|_{L_s^2(0,t)} \\ &\leq C \sum_{j=1}^p \|\langle s \rangle^{1+\delta} \partial_t u_j(s, 0)\|_{L_s^2(0,t)} \|\partial_x u_i(s, 0)\|_{L_s^\infty(0,t)} \\ &\quad + C \sum_{j,k=1}^p \|\langle s \rangle^{1+\delta} \partial_t u_j(s, 0)\|_{L_s^2(0,t)} \|u_k(s, 0)\|_{L_s^\infty(0,t)} \end{aligned}$$

$$\leq C \sum_{j=1}^p \|\langle s \rangle^{1+\delta} \partial_t u_j(s, 0)\|_{L_s^2(0,t)} \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)). \quad (2.39)$$

Thus we can obtain

$$P^+(\partial_x \partial_t u(t)) \leq C P_1^-(u(t)). \quad (2.40)$$

From (1.1), we also have that for $i = p+1, \dots, n$,

$$\begin{aligned} (\lambda_i - \sum_{j=1}^n \lambda_{ij} u_j) \partial_x^2 u_i &= -\partial_x \partial_t u_i + \sum_{j=1}^n \lambda_{ij} \partial_x u_j \partial_x u_i \\ &+ \sum_{j,k=1}^n \lambda_{ijk} u_j \partial_x u_k + \sum_{j,k=1}^n \lambda_{ijk} \partial_x u_j u_k. \end{aligned} \quad (2.41)$$

In view of (1.2), and noting that $|u|$ is sufficiently small, we see

$$\begin{aligned} |\partial_x^2 u_i| &\leq C |\partial_x \partial_t u_i| + C \sum_{j=1}^n |\lambda_{ij} \partial_x u_j \partial_x u_i| \\ &+ C \sum_{j,k=1}^n |\lambda_{ijk} u_j \partial_x u_k| + C \sum_{j,k=1}^n |\lambda_{ijk} \partial_x u_j u_k|, \end{aligned} \quad (2.42)$$

Thus we have

$$\begin{aligned} |\partial_x^2 u_i(t, 0)| &\leq C |\partial_x \partial_t u_i(t, 0)| + C \sum_{j=1}^n |\partial_x u_j(t, 0) \partial_x u_i(t, 0)| \\ &+ C \sum_{j,k=1}^n |\partial_x u_j(t, 0) u_k(t, 0)|, \end{aligned} \quad (2.43)$$

which implies

$$\begin{aligned} &\|\langle s \rangle^{1+\delta} \partial_x^2 u_i(t, 0)\|_{L_s^2(0,t)} \\ &\leq C \|\langle s \rangle^{1+\delta} \partial_x \partial_t u_i(t, 0)\|_{L_s^2(0,t)} + C \sum_{j=1}^n \|\langle s \rangle^{1+\delta} \partial_x u_j(t, 0)\|_{L_s^2(0,t)} \|\partial_x u_i(t, 0)\|_{L_s^\infty(0,t)} \\ &+ C \sum_{j,k=1}^n \|\langle s \rangle^{1+\delta} \partial_x u_j(t, 0)\|_{L_s^2(0,t)} \|u_k(t, 0)\|_{L_s^\infty(0,t)} \\ &\leq C \|\langle s \rangle^{1+\delta} \partial_x \partial_t u_i(t, 0)\|_{L_s^2(0,t)} + C \sum_{j=1}^n \|\langle s \rangle^{1+\delta} \partial_x u_j(t, 0)\|_{L_s^2(0,t)} \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)). \end{aligned} \quad (2.44)$$

It follows from (2.44) and (2.39) that

$$P^+(\partial_x^2 u(t)) \leq C P_1^-(u(t)) + C(P_1^-(u(t)) + P_1^+(u(t))) \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)). \quad (2.45)$$

In view of (2.30), (2.35), (2.40) and (2.45), we have

$$P_2^+(u(t)) \leq C P_2^-(u(t)) + C P_2^+(u(t)) \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \leq C P_2^-(u(t)) + C \epsilon_2 P_2^+(u(t)). \quad (2.46)$$

If ϵ_2 is sufficiently small, we can get (2.29). $\square$

3 Proof of Theorem 1.1

Now we will prove Theorem 1.1 by some bootstrap argument. Assume that u is a classical solution to the initial-boundary value problem (1.1), (1.5), (1.6). We will prove that there exist positive constants ϵ_0 and A such that

$$\sup_{0 \leq s \leq t} E_2(u(s)) + \mathcal{E}_2(u(t)) \leq A^2 \epsilon^2 \quad (3.1)$$

under the assumption

$$\sup_{0 \leq s \leq t} E_2(u(s)) + \mathcal{E}_2(u(t)) \leq 4A^2 \epsilon^2, \quad (3.2)$$

where $0 < \epsilon \leq \epsilon_0$.

For $l = 0, 1, 2$, Leibniz's rule gives

$$\partial_t(\partial_{t,x}^l u_i) + \lambda_i \partial_x(\partial_{t,x}^l u_i) = \sum_{j=1}^n \lambda_{ij} u_j \partial_x(\partial_{t,x}^l u_i) + G_i, \quad (3.3)$$

where

$$G_i = \sum_{j \neq i} \sum_{\substack{l_1+l_2=l \\ l_2 < l}} \lambda_{ij} c_{l_1 l_2} (\partial_{t,x}^{l_1} u_j) (\partial_x \partial_{t,x}^{l_2} u_i) + \sum_{j \neq k} \sum_{l_1+l_2=l} \lambda_{ijk} c_{l_1 l_2} (\partial_{t,x}^{l_1} u_j) (\partial_{t,x}^{l_2} u_k), \quad (3.4)$$

$c_{l_1 l_2}$ are some constants.

In view of (3.3), (1.2) and (1.9), it follows from Lemma 2.1 that

$$\begin{aligned} P_2^-(u(t)) &\leq C E_2(u(0)) + C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)) \\ &\quad + C \sum_{i=1}^p \sum_{l=0}^2 \int_0^t \|\langle x - \lambda_i s \rangle^{2+2\delta} \partial_{t,x}^l u_i G_i\|_{L_x^1(\mathbb{R}^+)} ds \end{aligned} \quad (3.5)$$

and

$$\begin{aligned} &\sup_{0 \leq s \leq t} E_2(u(s)) + \mathcal{E}_2(u(t)) \\ &\leq C E_2(u(0)) + C P_2^+(u(t)) + C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)) \\ &\quad + C \sum_{i=1}^n \sum_{l=0}^2 \int_0^t \|\langle x - \lambda_i s \rangle^{2+2\delta} \partial_{t,x}^l u_i G_i\|_{L_x^1(\mathbb{R}^+)} ds. \end{aligned} \quad (3.6)$$

While from (1.2) and Lemma 2.2, we have

$$P_2^+(u(t)) \leq C P_2^-(u(t)). \quad (3.7)$$

The combination of (3.6), (3.7) and (3.5) implies

$$\begin{aligned} &\sup_{0 \leq s \leq t} E_2(u(s)) + \mathcal{E}_2(u(t)) \\ &\leq C E_2(u(0)) + C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)) \\ &\quad + C \sum_{i=1}^n \sum_{l=0}^2 \int_0^t \|\langle x - \lambda_i s \rangle^{2+2\delta} \partial_{t,x}^l u_i G_i\|_{L_x^1(\mathbb{R}^+)} ds. \end{aligned} \quad (3.8)$$

By (3.4), (1.9), (1.10) and (1.11), we get

$$\begin{aligned}
& \sum_{i=1}^n \sum_{l=0}^2 \int_0^t \|\langle x - \lambda_i s \rangle^{2+2\delta} \partial_{t,x}^l u_i G_i\|_{L_x^1(\mathbb{R}^+)} ds \\
& \leq C \sum_{i=1}^n \sum_{j \neq i} \sum_{l=0}^2 \sum_{\substack{l_1+l_2=l \\ l_2 < l}} \|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^l u_i) (\partial_{t,x}^{l_1} u_j) (\partial_{t,x}^{l_2+1} u_i)\|_{L_{s,x}^1} \\
& + C \sum_{i=1}^p \sum_{l=0}^2 \sum_{\substack{l_1+l_2=l \\ l_2 < l}} \|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^l u_i) (\partial_{t,x}^{l_1} u_i) (\partial_{t,x}^{l_2+1} u_i)\|_{L_{s,x}^1} \\
& + C \sum_{i=1}^n \sum_{j \neq k} \sum_{l=0}^2 \sum_{l_1+l_2=l} \|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^l u_i) (\partial_{t,x}^{l_1} u_j) (\partial_{t,x}^{l_2} u_k)\|_{L_{s,x}^1} \\
& + C \sum_{i=1}^p \sum_{j=1}^i \sum_{l=0}^2 \sum_{l_1+l_2=l} \|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^l u_i) (\partial_{t,x}^{l_1} u_j) (\partial_{t,x}^{l_2} u_j)\|_{L_{s,x}^1} \\
& + C \sum_{i=p+1}^n \sum_{j=1}^p \sum_{l=0}^2 \sum_{l_1+l_2=l} \|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^l u_i) (\partial_{t,x}^{l_1} u_j) (\partial_{t,x}^{l_2} u_j)\|_{L_{s,x}^1} \\
& =: I + II + III + IV + V.
\end{aligned} \tag{3.9}$$

We first estimate I . We only consider the case $l = 2$. The case $l = 0, 1$ can be treated similarly. For $i = 1, 2, \dots, n$, $j \neq i$, $l = 2$, $l_1 + l_2 = 2$, $l_2 < 2$, when $l_1 = l_2 = 1$, it follows from Hölder inequality and Sobolev embedding $H^1(\mathbb{R}^+) \hookrightarrow L^\infty(\mathbb{R}^+)$ that

$$\begin{aligned}
& \|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^2 u_i) (\partial_{t,x} u_j) (\partial_{t,x} u_i)\|_{L_x^1} \\
& \leq \|\langle x - \lambda_j s \rangle^{\frac{-1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2}^2 \|\langle x - \lambda_j s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^\infty} \\
& \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)).
\end{aligned} \tag{3.10}$$

When $l_1 = 2$, $l_2 = 0$, it also follows from Hölder inequality and Sobolev embedding $H^1(\mathbb{R}^+) \hookrightarrow L^\infty(\mathbb{R}^+)$ that

$$\begin{aligned}
& \|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^2 u_i) (\partial_{t,x}^2 u_j) (\partial_{t,x} u_i)\|_{L_{s,x}^1} \\
& \leq \|\langle x - \lambda_j s \rangle^{\frac{-1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_j s \rangle^{\frac{-1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x} u_i\|_{L_s^\infty L_x^\infty} \\
& \quad \cdot \|\langle x - \lambda_j s \rangle^{1+\delta} \partial_{t,x}^2 u_j\|_{L_s^\infty L_x^2} \\
& \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)).
\end{aligned} \tag{3.11}$$

Thus we can get

$$I \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)). \tag{3.12}$$

Then we estimate II . For $i = 1, \dots, p$, $l = 2$, $l_1 + l_2 = 2$, $l_2 < 2$, when $l_1 = l_2 = 1$, from Hölder inequality, (1.8) and Sobolev embedding $H^1(\mathbb{R}^+) \hookrightarrow L^\infty(\mathbb{R}^+)$, we can obtain

$$\begin{aligned}
& \|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^2 u_i) (\partial_{t,x} u_i) (\partial_{t,x} u_i)\|_{L_{s,x}^1} \\
& \leq \|\langle x - \lambda_n s \rangle^{\frac{-1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2}^2 \|\langle x - \lambda_n s \rangle^{1+\delta} \partial_{t,x} u_i\|_{L_{s,x}^\infty} \\
& \leq C \|\langle x - \lambda_n s \rangle^{\frac{-1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2}^2 \|\langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x} u_i\|_{L_{s,x}^\infty}
\end{aligned}$$

$$\leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)). \quad (3.13)$$

The case $l_1 = 2, l_2 = 0$ can be treated by the same way. Thus we can get

$$II \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)). \quad (3.14)$$

Now we consider *III*. Fix $i = 1, 2, \dots, n, j \neq k, l = 2, l_1 + l_2 = 2$. When $l_1 = l_2 = 1$, if $j = i \neq k$, we see

$$\begin{aligned} & \|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^2 u_i) (\partial_{t,x} u_i) (\partial_{t,x} u_k)\|_{L_{s,x}^1} \\ & \leq \|\langle x - \lambda_k s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_k s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x} u_i\|_{L_{s,x}^2} \\ & \quad \cdot \|\langle x - \lambda_k s \rangle^{1+\delta} \partial_{t,x} u_k\|_{L_{s,x}^\infty} \\ & \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)); \end{aligned} \quad (3.15)$$

if $k = i \neq j$, similarly we also have

$$\begin{aligned} & \|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^2 u_i) (\partial_{t,x} u_j) (\partial_{t,x} u_i)\|_{L_{s,x}^1} \\ & \leq \|\langle x - \lambda_j s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_j s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x} u_i\|_{L_{s,x}^2} \\ & \quad \cdot \|\langle x - \lambda_j s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^\infty} \\ & \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)); \end{aligned} \quad (3.16)$$

if $j \neq k, j \neq i, k \neq i$, noting

$$\begin{aligned} |x - \lambda_i t| & \leq |x - \lambda_j t| + |\lambda_j - \lambda_i| t \\ & \leq |x - \lambda_j t| + C |\lambda_j - \lambda_k| t \leq C (|x - \lambda_j t| + |x - \lambda_k t|), \end{aligned} \quad (3.17)$$

which implies

$$\langle x - \lambda_i t \rangle^{1+\delta} \leq C (\langle x - \lambda_j t \rangle^{1+\delta} + \langle x - \lambda_k t \rangle^{1+\delta}), \quad (3.18)$$

by (3.18) we have

$$\begin{aligned} & \|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^2 u_i) (\partial_{t,x} u_j) (\partial_{t,x} u_k)\|_{L_{s,x}^1} \\ & \leq C \|\langle x - \lambda_i s \rangle^{1+\delta} (\partial_{t,x}^2 u_i) \langle x - \lambda_j s \rangle^{1+\delta} (\partial_{t,x} u_j) (\partial_{t,x} u_k)\|_{L_{s,x}^1} \\ & \quad + C \|\langle x - \lambda_i s \rangle^{1+\delta} (\partial_{t,x}^2 u_i) (\partial_{t,x} u_j) \langle x - \lambda_k s \rangle^{1+\delta} (\partial_{t,x} u_k)\|_{L_{s,x}^1} \\ & \leq C \|\langle x - \lambda_k s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_k s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_j s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^2} \\ & \quad \cdot \|\langle x - \lambda_k s \rangle^{1+\delta} \partial_{t,x} u_k\|_{L_{s,x}^\infty} \\ & \quad + C \|\langle x - \lambda_j s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_j s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_k s \rangle^{1+\delta} \partial_{t,x} u_k\|_{L_{s,x}^2} \\ & \quad \cdot \|\langle x - \lambda_j s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^\infty} \\ & \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)). \end{aligned} \quad (3.19)$$

The case $l_1 = 2, l_2 = 0$ can be treated similarly. When $l_1 = 0, l_2 = 2$, if $j = i \neq k$, we see

$$\|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^2 u_i) (u_i) (\partial_{t,x}^2 u_k)\|_{L_{s,x}^1}$$

$$\begin{aligned}
&\leq \|\langle x - \lambda_k s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_k s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} u_i\|_{L_s^2 L_x^\infty} \\
&\quad \cdot \|\langle x - \lambda_k s \rangle^{1+\delta} \partial_{t,x}^2 u_k\|_{L_s^\infty L_x^2} \\
&\leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t));
\end{aligned} \tag{3.20}$$

if $k = i \neq j$, we have

$$\begin{aligned}
&\|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^2 u_i) u_j (\partial_{t,x}^2 u_i)\|_{L_{s,x}^1} \\
&\leq \|\langle x - \lambda_j s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2}^2 \|\langle x - \lambda_j s \rangle^{1+\delta} u_j\|_{L_{s,x}^\infty} \\
&\leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t));
\end{aligned} \tag{3.21}$$

if $j \neq k, j \neq i, k \neq i$, by (3.18) we get

$$\begin{aligned}
&\|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^2 u_i) u_j (\partial_{t,x}^2 u_k)\|_{L_{s,x}^1} \\
&\leq C \|\langle x - \lambda_i s \rangle^{1+\delta} (\partial_{t,x}^2 u_i) \langle x - \lambda_j s \rangle^{1+\delta} u_j (\partial_{t,x}^2 u_k)\|_{L_{s,x}^1} \\
&\quad + C \|\langle x - \lambda_i s \rangle^{1+\delta} (\partial_{t,x}^2 u_i) u_j \langle x - \lambda_k s \rangle^{1+\delta} (\partial_{t,x}^2 u_k)\|_{L_{s,x}^1} \\
&\leq C \|\langle x - \lambda_k s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_k s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_j s \rangle^{1+\delta} u_j\|_{L_s^2 L_x^\infty} \\
&\quad \cdot \|\langle x - \lambda_k s \rangle^{1+\delta} \partial_{t,x}^2 u_k\|_{L_s^\infty L_x^2} \\
&\quad + C \|\langle x - \lambda_j s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_j s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_k s \rangle^{1+\delta} \partial_{t,x}^2 u_k\|_{L_{s,x}^2} \\
&\quad \cdot \|\langle x - \lambda_j s \rangle^{1+\delta} u_j\|_{L_{s,x}^\infty} \\
&\leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)).
\end{aligned} \tag{3.22}$$

Thus we have

$$III \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)). \tag{3.23}$$

Now we consider *IV*. For $i = 1, \dots, p, j = 1, \dots, i, l = 2, l_1 + l_2 = 2$, if $l_1 = l_2 = 1$, by (1.8) we have

$$\begin{aligned}
&\|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^2 u_i) (\partial_{t,x} u_j) (\partial_{t,x} u_j)\|_{L_{s,x}^1} \\
&\leq \|\langle x - \lambda_n s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_n s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^2} \\
&\quad \cdot \|\langle x - \lambda_n s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^\infty} \\
&\leq c \|\langle x - \lambda_n s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_n s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_j s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^2} \\
&\quad \cdot \|\langle x - \lambda_j s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^\infty} \\
&\leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)).
\end{aligned} \tag{3.24}$$

The case $l_1 = 2, l_2 = 0$ and the case $l_1 = 0, l_2 = 2$ can be treated by the same way. Thus we can get

$$IV \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)). \tag{3.25}$$

Finally, we consider *V*. For $i = p+1, \dots, n, j = 1, \dots, p, l = 2, l_1 = l_2 = 1$, by (1.8) we have

$$\|\langle x - \lambda_i s \rangle^{2+2\delta} (\partial_{t,x}^2 u_i) (\partial_{t,x} u_j) (\partial_{t,x} u_j)\|_{L_{s,x}^1}$$

$$\begin{aligned}
&\leq \|\langle x - \lambda_p s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_p s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^2} \\
&\quad \cdot \|\langle x - \lambda_p s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^\infty} \\
&\leq \|\langle x - \lambda_p s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_i s \rangle^{1+\delta} \partial_{t,x}^2 u_i\|_{L_{s,x}^2} \|\langle x - \lambda_p s \rangle^{-\frac{1+\delta}{2}} \langle x - \lambda_j s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^2} \\
&\quad \cdot \|\langle x - \lambda_j s \rangle^{1+\delta} \partial_{t,x} u_j\|_{L_{s,x}^\infty} \\
&\leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)). \tag{3.26}
\end{aligned}$$

The case $l_1 = 2, l_2 = 0$ and the case $l_1 = 0, l_2 = 2$ can be treated by the same way. Thus we can see

$$V \leq C \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)). \tag{3.27}$$

From (3.8), (3.9), (3.12), (3.14), (3.23), (3.25) and (3.27), we have

$$\sup_{0 \leq s \leq t} E_2(u(s)) + \mathcal{E}_2(u(t)) \leq C_1 E_2(u(0)) + C_1 \sup_{0 \leq s \leq t} E_2^{1/2}(u(s)) \mathcal{E}_2(u(t)). \tag{3.28}$$

Under the assumption (3.2), we have

$$\sup_{0 \leq s \leq t} E_2(u(s)) + \mathcal{E}_2(u(t)) \leq C_0 \epsilon^2 + 8C_1 A^3 \epsilon^3. \tag{3.29}$$

Taking $A^2 = 2C_0$ and ϵ_0 so small that

$$16C_1 A \epsilon_0 \leq 1, \tag{3.30}$$

for any ϵ with $0 < \epsilon \leq \epsilon_0$, we have

$$\sup_{0 \leq s \leq t} E_2(u(s)) + \mathcal{E}_2(u(t)) \leq A^2 \epsilon^2, \tag{3.31}$$

which completes the proof of Theorem 1.1.

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